S2 Hierarchical clustering

S2.1 Partition overlap with the consensus partition

In the pipeline of our analysis, an important step is the use of the *multilayer hierarchical stochastic blockmodel* (or hSBM). We run 100 iterations of the algorithm and derive the consensus partition to increase robustness and generalizability of our results. This approach helps to account for the inherent stochasticity and variability in the algorithm, ensuring that the identified partition is not an artifact of a single run but a consistent representation of the data.

In Fig. S2, we present the maximum overlap and the Normalized Mutual Information (NMI) between the document and word partitions obtained from different single iterations of the multilayer hSBM, compared to the consensus partition. The panels (a) and (b) represent the maximum overlap, while panels (c) and (d) show the NMI. Moreover, panels (a) and (c) represent the document clusters, while panels (b) and (d) refer to the word clusters. The values in each cell (i, j) of the panel indicate the average overlap (with standard deviation among parentheses) for the single partitions at level i and the consensus partition at level j. This provides a comprehensive view of the agreement between individual iterations and the consensus partition. Moreover, the number of clusters at the various levels is displayed in the figure.

One notable finding is the high average overlap between the single partitions and the consensus partition, particularly at specific levels. This indicates a strong degree of agreement among the different iterations and the consensus, suggesting robustness and reliability in the identified partitions. Less agreement is instead found in the low levels, which have a higher level of granularity. For example, at level 0 the partitions have more than 500 clusters.

Furthermore, the NMI values provide insights into the information shared between the individual partitions and the consensus one. Higher NMI values indicate a greater level of similarity and shared information. The observed NMI values demonstrate a significant degree of agreement between the single partitions and the consensus partition, further supporting the reliability of the identified partitions.

Overall, these results highlight the effectiveness of the consensus partition in capturing the shared structure across multiple iterations of the hSBM. The high levels of overlap and NMI indicate a consistent and reliable identification of partitions, contributing to a more accurate understanding of the underlying patterns and structures within the analyzed data.

It should be noted that these results are specific to the current dataset and methodology employed. Further investigations and comparisons with alternative approaches are necessary to validate and generalize these findings. Nonetheless, the observed agreement between the single iterations and the consensus partition provides a promising avenue for future research in network analysis and partitioning algorithms.

S2.2 Keywords annotation

As described in the main text, the hSBM algorithm produces a hierarchical clustering of both articles and words (i.e. topics). Here we give more details on the procedure we used to manually assign keywords to represent the different document clusters. First, for each cluster at level 4, we manually inspect the most frequent words in the publications' titles. We then look at the most significant topics represented in the articles, as quantified by the normalized mixture proportion [21]. For instance, as we can see

depicted in Fig. S3(b) for the Blockchain cluster, or in Fig. S4(b) for the Governance cluster, we can visualize the hierarchical tree of topics, highlighting the top 10 most significant ones. We have also reported in the tree the significance level (> 1) with the * symbol. Moreover, for each of these topic,

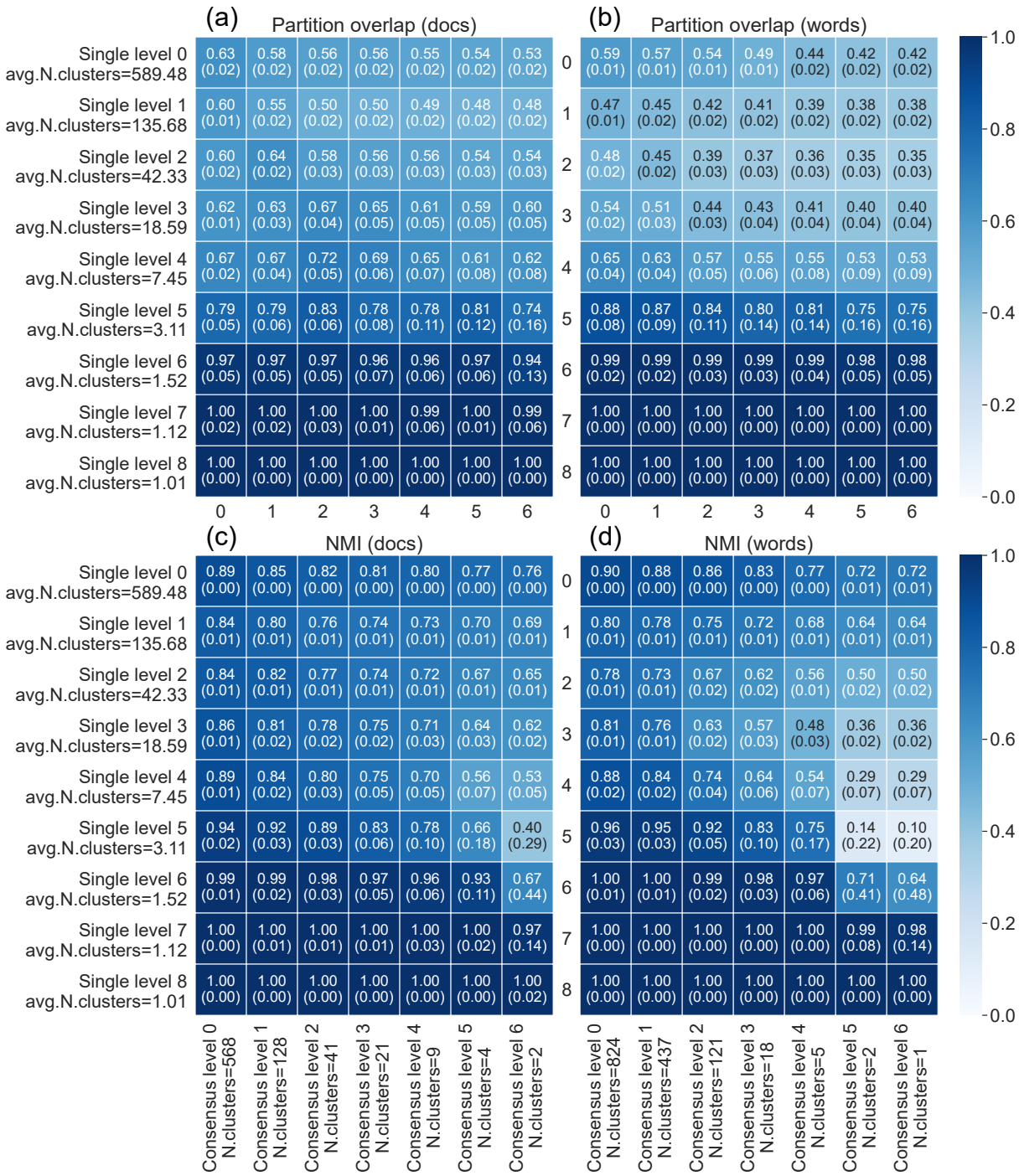

Figure S2: **Overlap between single iterations of the hSBM with the consensus partitions.** Maximum overlap **(a-b)** and Normalized Mutual Information **(c-d)** between the document **(a,c)** and word **(b,d)** partitions obtained from the different single iterations of the multilayer hSBM versus the consensus partition. Such consensus partition ˆis obtained by maximizing the sum of overlaps with all the partitions. The value of a cell in row i and column j in the panel refers to average (and standard deviation within parentheses) for the single partitions at level i and the consensus partition at level j.

the top 20 most significant words are printed. We also look at the fields of study represented in each cluster in time, as shown in Fig. S3(a) for Blockchain, or in Fig. S4(a) for Governance. Finally, we look at the most important articles in the cluster, as represented by different metrics: overall number of citations, number of citations within the *(de)centralization* dataset, highest knowledge flows towards other clusters, and most central publications according to different network centrality measures (degree, pagerank, betweenness, closeness, katz).

For clarity, we have then chosen one representative keyword for each cluster at the 3rd level. In Table S4, we show the results of the keyword annotation procedure. Starting from the leftmost column, we show each cluster at level 3 with its representative keyword and the number of publications in it, to then show the annotated keywords in each cluster at level 4 (and the number of publications) present in its hierarchical branch in separate rows. Clusters with less than 500 documents have been disregarded.

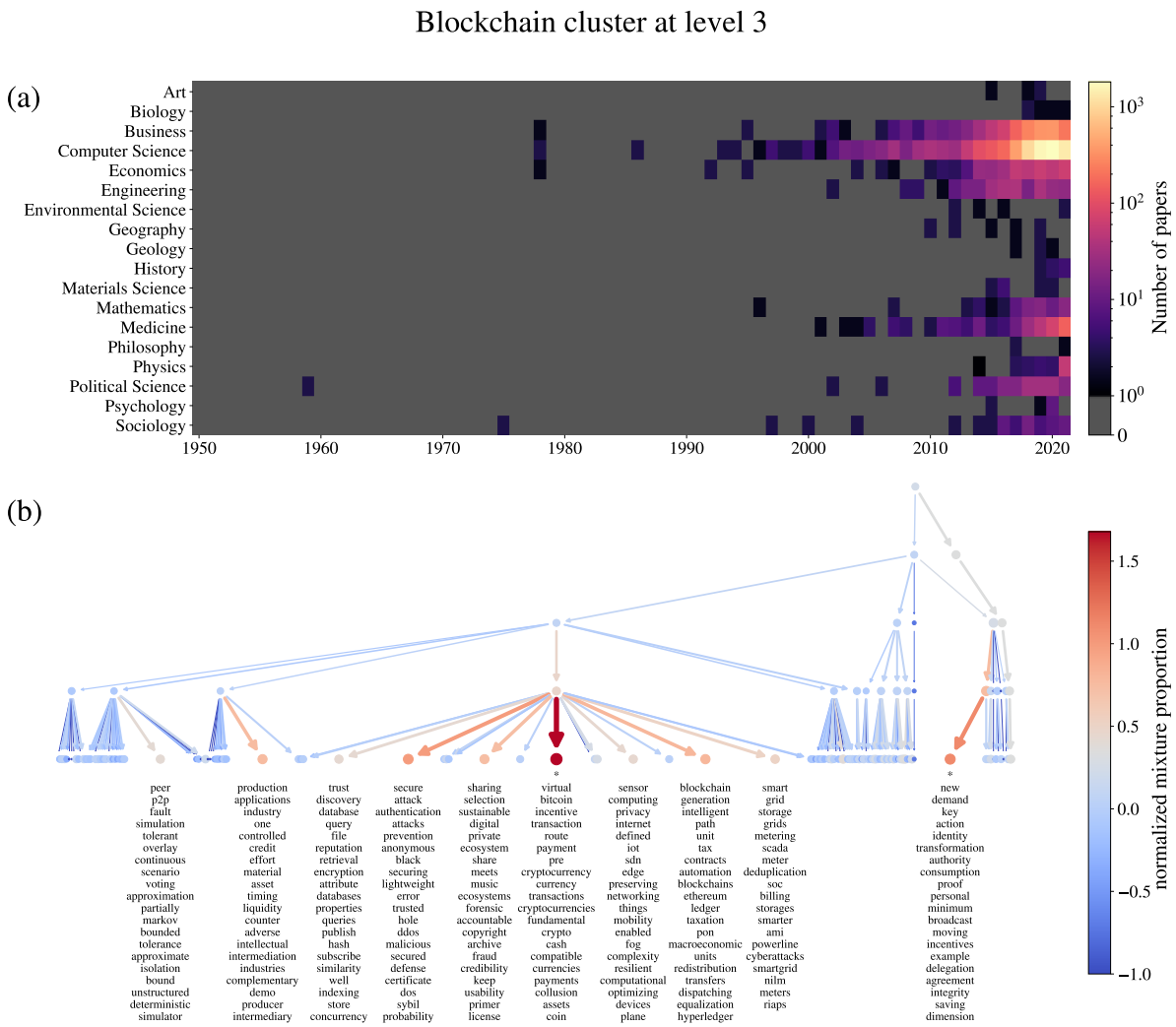

Figure S3: **Keywords annotation for Blockchain.** **(a)** Number of papers in time in each field of study for the Blockchain cluster. **(b)** Hierarchical topic tree, highlighting the top 20 words of the 10 most significant topics in the Blockchain cluster according to the normalized mixture proportion (in color). This information is used to aid the keywords annotation manual procedure.